Bo-Yu Chen

Researcher and builder at the intersection of MIR, DSP, Instrument Making, and HCI.

EDUCATION

National Taiwan University (NTU)
Sep 2021 – Aug 2025

M.S. in Networking and Multimedia
Taipei, Taiwan

National Taiwan University of Science and Technology (NTUST)
Sep 2013 – Jun 2018

B.S. in Electrical Engineering
Taipei, Taiwan

WORK EXPERIENCE

Rhythm Culture Corporation
Jul 2023 – Present

Technical Lead | Taiwan

- Developed TMC-CL1, a web-based music sampler for high school classrooms.
- Built agent-based Music AI systems spanning DJ, Producer, and Tutor roles for interactive creation and feedback workflows.
- Led projects funded by Taipei Music Center, TAICCA, and Taipei Government.
- Recognized at NeurIPS Creative AI Track 2024 and ISMIR LLM4Music Workshop 2025.

Zone Sound Creative Corporation
Mar 2019 – Aug 2019

Interactive Engineer | Taiwan

- Shipped "Taptap," a browser-based performance interface using a cappella samples; exhibited at Vocal Asia 2019 (Japan).
- Produced "FusionTaiwanBeat," an audio-visual system featuring Taiwanese electronic music with DJ Sonica Calico & VJ Celso Urro; exhibited at MIDEM 2019 (France).
- Developed "Duplex Transmission," a real-time audience-performer interaction system; performed at Songshan Culture Park (Taiwan) with Po-Hao Chi.

RESEARCH EXPERIENCE

Computational Physicality Lab, NTU
Sep 2021 – Aug 2025

Research Assistant | Taiwan

- Developed FlueBricks, a modular toolkit for designing customizable flute-like instruments.
- Utilized digital fabrication (Fusion 360, 3D printing) for rapid prototyping.
- Engineered Python & Arduino-based evaluation systems for acoustic analysis.

Creative AI Lab, Sony Group Corporation

Nov 2020 – Sep 2021

Research Intern | Tokyo, Japan (Remote)

- Developed DJTransGAN, the first AI model for generating smooth DJ-like transitions.
- Designed automated DJ mixing systems integrating web crawlers and deep learning models.
- Open-sourced the project; 100+ GitHub stars.

Music and AI Lab, Academia Sinica

Jul 2019 – Sep 2023

Research Assistant | Taiwan

- Led research in intelligent music production, loop-based retrieval and generation.
- Published 5+ papers in top music technology conferences (ISMIR).
- Mentored 5+ junior researchers and developed DeepMIR course materials.
- Collaborated with TikTok, Sony, and MTG at UPF on joint publications.

TEACHING

BLND Creative Coding School

Aug 2020 – Jul 2023

Instructor | Taiwan

- Taught creative coding, AI art, and interactive device design with Arduino.
- Led courses on Stable Diffusion workflows and GAN-based ML image generation from data collection to training.

OrangeApple Children Programming School

Jul 2015 – Sep 2019

Instructor | Taiwan

- Taught web development and algorithms in JavaScript and C++ for elementary to senior high students.
- Covered advanced topics (sorting, recursion, dynamic programming, intro AI) and coached students for the APCS exam.

THESIS

FlueBricks: A Modular Toolkit for Interactive Flute-like Instrument Systems

Bo-Yu Chen, Chiao-Wei Huang, Lung-Pan Cheng

NTU, 2025

PUBLICATIONS

FlueBricks: A Construction Kit of Flute-like Instrument for Acoustic Reasoning

Bo-Yu Chen, Chiao-Wei Huang, Lung-Pan Cheng

CHI 2026, 2026 Acceptance rate: 25%

AI TrackMate: Finally, Someone Who Will Give Your Music More Than Just "Sounds Great!"

Yi-Lin Jiang, Chia-Ho Hsiung, Yen-Tung Yeh, Lu-Rong Chen, ***Bo-Yu Chen***

NeurIPS Creative AI Track, 2024 Acceptance rate: 31%

Automatic DJ Transitions with Differentiable Audio Effects and Generative Adversarial Networks

Bo-Yu Chen, Wei-Han Hsu, Wei-Hsiang Liao, Marco A. Martínez Ramírez, Yuki Mitsufuji, Yi-Hsuan Yang

ICASSP, 2022 Acceptance rate: 47%

Exploiting Pre-trained Feature Networks for Generative Adversarial Networks in Audio-Domain Loop Generation

Yen-Tung Yeh, ***Bo-Yu Chen***, Yi-Hsuan Yang

ISMIR, 2022 Acceptance rate: 43%

A Benchmarking Initiative for Audio-Domain Music Generation Using the Freesound Loop Dataset

Tun-Min Hung, ***Bo-Yu Chen***, Yen-Tung Yeh, Yi-Hsuan Yang

ISMIR, 2021 Acceptance rate: 34%

Neural Loop Combiner: Neural Network Models for Assessing the Compatibility of Loops

Bo-Yu Chen, Jordan Smith, Yi-Hsuan Yang

ISMIR, 2020 Acceptance rate: 38%

AWARDS

FlueBricks | TAICHI 2025 Best Paper Award 🏆

REVIEWER

ISMIR | 2020–2025

ICASSP | 2025

NeurIPS Creative AI Track | 2025